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Brown Planthopper

Recognize the problem

The brown planthopper is a major problem on rice, particularly in low land. In Tamil Nadu, it is known as "Pugaiyan". The adult has a brown body and chestnut brown eyes. Two forms of panthopper, namely long winged and short winged, are seen. As a result of feeding by both nymphs and adults at the base of the tillers, plants turn yellow and dry up rapidly. This condition is called 'hopper burn', in which circular patches of drying and lodging of matured plants is seen. If timely control measures are not adopted it may cause up to 60% crop loss.

Brown plant hopper adult. (Photo by IRRI)



Background

The insect infests the rice crop at all stages of plant growth. Nymphs and adults congregate at the base of the plant above the water level. They damage rice directly through feeding and also by transmitting two viruses, rice ragged stunt virus and rice grassy stunt virus. Excessive use of urea and certain chemicals can lead to outbreak by increasing the egg laying of the brown plant hopper and by reducing populations of natural enemies.

Hopper burn. (Photo by forestryimages.org)



Management

- Grow resistant/ tolerant varieties like ADT 36, Co 46, Co 42, IR 36, Aruna
- Provide 30 cm rogue spacing at every 2.5 m to reduce the pest incidence. Avoid the use of excessive nitrogenous fertilizers. Control irrigation by intermittent draining. Grow Azolla to increase wolf spider population, which will control BPH.
- Set up light traps @ 5 / acre between 7-9 pm. Install 20 yellow pan traps during the day time
- Conserve natural enemies like wolf spider, mirid bugs etc.
- Avoid chemicals like fenthion and quinalphos which can cause resurgence of the pest
- If the pest persists after adopting the above management practices then chemical control may be necessary. Drain water from the field then spray any one of the following chemicals: buprofezin 25% EC @ 625 ml/ha, imidacloprid 17.8 SL @ 100 ml/ha, thiamethoxam 20 % WDG @ 100 g/ha, neem oil 3% @ 15 lit/ha or 25 kg of neem seed kernel powder in 600 litres of water for 1 ha.

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > ***Nilaparvata lugens***

The recommendations in this factsheet are relevant to: India

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